UNIT 7

Redox Reactions

Where there is oxidation, there is always reduction – Chemistry is essentially a study of redox systems.

What this unit sets out to do

- identify redox reactions as a class of reactions in which oxidation and reduction reactions occur simultaneously
- define the terms oxidation, reduction, oxidant (oxidising agent) and reductant (reducing agent), and explain the mechanism of redox reactions by electron transfer
- use the concept of oxidation number to identify oxidant and reductant in a reaction
- classify redox reaction into combination (synthesis), decomposition, displacement and disproportionation reactions, and suggest a comparative order among various reductants and oxidants
- balance chemical equations using (i) oxidation number (ii) half reaction method, and learn the concept of redox reactions in terms of electrode processes

Where redox reactions turn up

- chemistry deals with varieties of matter and change of one kind of matter into the other, transformation occurring through the various types of reactions, and one important category of such reactions is redox reactions
- a number of phenomena, both physical as well as biological, are concerned with redox reactions, and these reactions find extensive use in pharmaceutical, biological, industrial, metallurgical and agricultural areas
- burning fuels for domestic, transport and other commercial purposes, electrochemical processes for extracting highly reactive metals and non-metals, manufacturing compounds like caustic soda, the operation of dry and wet batteries and the corrosion of metals all fall within the purview of redox processes
- of late, environmental issues like Hydrogen Economy, the use of liquid hydrogen as fuel, and the development of the Ozone Hole have started figuring under redox phenomenon

7.1 Oxidation, as it was first defined

- originally the term oxidation was used to describe the addition of oxygen to an element or a compound, and because of the presence of dioxygen in the atmosphere at about 20 per cent many elements combine with it, which is the principal reason they commonly occur on the earth in the form of their oxides
- on that limited definition two magnesium plus dioxygen giving two magnesium oxide, and sulphur plus dioxygen giving sulphur dioxide, are both oxidation processes

Widening the term oxidation

From adding oxygen to removing hydrogen

- in the two reactions of the previous page the elements magnesium and sulphur are oxidised on account of addition of oxygen to them, and similarly methane is oxidised owing to the addition of oxygen to it
- a careful examination of that methane reaction, in which hydrogen has been replaced by oxygen, prompted chemists to reinterpret oxidation in terms of removal of hydrogen, and so the scope of the term was broadened to include the removal of hydrogen from a substance, as when two hydrogen sulphide and dioxygen give two sulphur and two water
- as knowledge grew it was natural to extend the term to reactions which do not involve oxygen but other electronegative elements, magnesium reacting with fluorine, chlorine and sulphur to give magnesium fluoride, magnesium chloride and magnesium sulphide

And on to removing an electropositive element

- incorporating those three reactions within the fold of oxidation encouraged chemists to consider not only the removal of hydrogen as oxidation but also the removal of electropositive elements
- the reaction of potassium ferrocyanide with hydrogen peroxide to give potassium ferricyanide and potassium hydroxide is interpreted as oxidation due to the removal of the electropositive element potassium
- to summarise, the term oxidation is defined as the addition of oxygen or an electronegative element to a substance, or the removal of hydrogen or an electropositive element from a substance

Reduction, widened the same way, and the word redox

- in the beginning reduction was considered as removal of oxygen from a compound, but the term has been broadened these days to include removal of oxygen or an electronegative element from a substance, or addition of hydrogen or an electropositive element to a substance
- the examples given are two mercuric oxide giving two mercury and dioxygen, which removes oxygen from mercuric oxide, two ferric chloride with dihydrogen giving two ferrous chloride and two hydrogen chloride, which removes the electronegative element chlorine, ethene with dihydrogen giving ethane, which adds hydrogen, and two mercuric chloride with stannous chloride giving mercurous chloride and stannic chloride, which adds mercury
- in that last reaction simultaneous oxidation of stannous chloride to stannic chloride is also occurring because of the addition of the electronegative element chlorine to it, and it was soon realised that oxidation and reduction always occur simultaneously, hence the word redox was coined for this class of chemical reactions

Problem 7.1, which species is oxidised and which reduced

- the question gives three reactions and asks which species undergoes oxidation and which reduction, (i) hydrogen sulphide with chlorine giving two hydrogen chloride and sulphur, (ii) three ferrous ferric oxide with eight aluminium giving nine iron and four aluminium oxide, and (iii) two sodium with dihydrogen giving two sodium hydride
- in (i) hydrogen sulphide is oxidised because a more electronegative element, chlorine, is added to hydrogen, or a more electropositive element, hydrogen, has been removed from sulphur, and chlorine is reduced due to addition of hydrogen to it
- in (ii) aluminium is oxidised because oxygen is added to it, and the solution continues overleaf

Problem 7.1, the rest of the solution

- ferrous ferric oxide is reduced because oxygen has been removed from it
- in (iii), with the careful application of the concept of electronegativity only, we may infer that sodium is oxidised and hydrogen is reduced
- reaction (iii) chosen here prompts us to think in terms of another way to define redox reactions

7.2 Redox reactions in terms of electron transfer reactions

Why the ionic products suggest a rewrite

- two sodium with chlorine giving two sodium chloride, four sodium with dioxygen giving two sodium oxide, and two sodium with sulphur giving sodium sulphide are redox reactions because in each of them sodium is oxidised due to the addition of either oxygen or a more electronegative element, while chlorine, oxygen and sulphur are reduced because to each of these the electropositive element sodium has been added
- from our knowledge of chemical bonding we also know that sodium chloride, sodium oxide and sodium sulphide are ionic compounds and perhaps better written with the charges shown, and the development of charges on the species produced suggests we rewrite the three reactions in that manner

Half reactions, and the four definitions they give

- for convenience each of those processes can be considered as two separate steps, one involving the loss of electrons and the other the gain of electrons, so the formation of sodium chloride is two sodium giving two sodium ion plus two electrons, and chlorine plus two electrons giving two chloride ion
- each of those steps is called a half reaction, which explicitly shows involvement of electrons, and the sum of the half reactions gives the overall reaction
- half reactions that involve loss of electrons are called oxidation reactions and those that involve gain of electrons are called reduction reactions, this new way of defining them achieved only by establishing a correlation between the behaviour of species as per the classical idea and their interplay in electron-transfer change
- sodium, which is oxidised, acts as a reducing agent because it donates electron to each of the elements interacting with it and thus helps in reducing them, while chlorine, oxygen and sulphur are reduced and act as oxidising agents because these accept electrons

The four terms, stated as electron book-keeping

- oxidation is loss of electron or electrons by any species
- reduction is gain of electron or electrons by any species
- an oxidising agent is an acceptor of electron or electrons
- a reducing agent is a donor of electron or electrons

Problem 7.2, sodium hydride as a redox change

- the question asks us to justify that two sodium with dihydrogen giving two sodium hydride is a redox change
- since the compound formed is an ionic compound, which may also be represented with a sodium cation and a hydride anion, this suggests that one half reaction in this process is two sodium giving two sodium ion plus two electrons, and the solution continues overleaf

A strip of metallic zinc standing in a beaker of aqueous copper nitrate

The strip becoming coated with reddish metallic copper

The blue colour of the solution fading as the copper ions leave it

Figure 7.A Redox reaction between zinc and aqueous solution of copper nitrate occurring in a beaker.

Problem 7.2, the rest of the solution

- the other half reaction is dihydrogen plus two electrons giving two hydride ion
- this splitting of the reaction under examination into two half reactions automatically reveals that here sodium is oxidised and hydrogen is reduced, therefore the complete reaction is a redox change

7.2.1 Competitive electron transfer reactions

Zinc in copper nitrate, and what settles out

- place a strip of metallic zinc in an aqueous solution of copper nitrate for about one hour and you may notice that the strip becomes coated with reddish metallic copper and the blue colour of the solution disappears
- formation of zinc ions among the products can easily be judged when the blue colour due to the copper ion has disappeared, and if hydrogen sulphide gas is passed through the colourless solution containing zinc ions, appearance of white zinc sulphide can be seen on making the solution alkaline with ammonia
- in the reaction zinc has lost electrons to form the zinc ion and therefore zinc is oxidised, and evidently if zinc is oxidised, releasing electrons, something must be reduced, accepting the electrons lost by zinc, so copper ion is reduced by gaining them

Where the equilibrium sits, and two more couples

- to investigate the state of equilibrium, place a strip of metallic copper in a zinc sulphate solution, where no visible reaction is noticed and the attempt to detect copper ions by passing hydrogen sulphide gas to produce the black colour of cupric sulphide does not succeed
- cupric sulphide has such a low solubility that this is an extremely sensitive test, yet the amount of copper ion formed cannot be detected, so we conclude that the state of equilibrium for that reaction greatly favours the products over the reactants
- extending the electron transfer reaction to copper metal and silver nitrate solution in water, the solution develops blue colour due to the formation of copper ions, copper is oxidised to the copper ion and the silver ion is reduced to silver, and equilibrium again greatly favours the products
- by way of contrast, compare the reaction of metallic cobalt placed in nickel sulphate solution, whose outcome is taken up on the next page

Two beakers, one of copper metal in aqueous silver nitrate

Silver depositing on the copper as the copper dissolves

The solution turning blue as copper ions enter it

Figure 7.B Redox reaction between copper and aqueous solution of silver nitrate occurring in a beaker.

Cobalt and nickel, where neither side wins

- at equilibrium chemical tests reveal that both the nickel ion and the cobalt ion are present at moderate concentrations, so in this case neither the reactants, cobalt and the nickel ion, nor the products, the cobalt ion and nickel, are greatly favoured
- this competition for release of electrons incidently reminds us of the competition for release of protons among acids, and the similarity suggests we might develop a table in which metals and their ions are listed on the basis of their tendency to release electrons, just as we do in the case of acids to indicate the strength of the acids
- by comparison we have come to know that zinc releases electrons to copper and copper releases electrons to silver, so the electron releasing tendency is zinc over copper over silver, and a longer list would be a metal activity series or electrochemical series
- the competition for electrons between various metals helps us design a class of cells named Galvanic cells, in which the chemical reactions become the source of electrical energy, studied further in Class XII

7.3 Oxidation number

A shift rather than a transfer

- a less obvious example of electron transfer is realised when hydrogen combines with oxygen to form water, where we can visualise the hydrogen atom going from a neutral zero state in dihydrogen to a positive state in water, and the oxygen atom going from a zero state in dioxygen to a dinegative state in water
- it is assumed that there is an electron transfer from hydrogen to oxygen and consequently dihydrogen is oxidised and dioxygen is reduced, however the charge transfer is only partial and is perhaps better described as an electron shift rather than a complete loss by hydrogen and gain by oxygen
- the same holds for a good number of other reactions involving covalent compounds, two such examples being dihydrogen with chlorine giving hydrogen chloride, and methane with four chlorine giving carbon tetrachloride and four hydrogen chloride

What oxidation number is for

- in order to keep track of electron shifts in reactions forming covalent compounds, a more practical method of using oxidation number has been developed, in which it is always assumed that there is a complete transfer of electron from a less electronegative atom to a more electonegative atom
- on that assumption the three reactions above are rewritten with a charge marked on each atom taking part, and it may be emphasised that the assumption of electron transfer is made for book-keeping purpose only
- oxidation number denotes the oxidation state of an element in a compound ascertained according to a set of rules formulated on the basis that the electron pair in a covalent bond belongs entirely to the more electronegative element

Why a set of rules is needed at all

- it is not always possible to remember or make out easily in a compound or ion which element is more electronegative than the other, therefore a set of rules has been formulated to determine the oxidation number of an element in a compound or ion
- if two or more than two atoms of an element are present in the molecule or ion, such as sodium thiosulphate or the dichromate ion, the oxidation number of the atom of that element will then be the average of the oxidation number of all the atoms of that element

Rules 1 to 3

- in elements in the free or the uncombined state each atom bears an oxidation number of zero, so each atom in dihydrogen, dioxygen, chlorine, ozone, white phosphorus, sulphur, sodium, magnesium and aluminium has the oxidation number zero
- for ions composed of only one atom the oxidation number is equal to the charge on the ion, and in their compounds all alkali metals have oxidation number of plus one, all alkaline earth metals plus two, and aluminium is regarded to have plus three in all its compounds
- the oxidation number of oxygen in most compounds is minus two, with two kinds of exception, one in peroxides such as hydrogen peroxide and sodium peroxide, where each oxygen atom is assigned minus one, and in superoxides such as potassium and rubidium superoxide, where each is assigned minus one half
- the second exception appears rarely, when oxygen is bonded to fluorine, and in oxygen difluoride and dioxygen difluoride the oxygen is assigned plus two and plus one respectively, the number depending on the bonding state of oxygen but now a positive figure only

Rules 4 to 6

- the oxidation number of hydrogen is plus one, except when it is bonded to metals in binary compounds, that is compounds containing two elements, and in lithium hydride, sodium hydride and calcium hydride its oxidation number is minus one
- in all its compounds fluorine has an oxidation number of minus one, other halogens also have minus one when they occur as halide ions in their compounds, and chlorine, bromine and iodine when combined with oxygen, for example in oxoacids and oxoanions, have positive oxidation numbers
- the algebraic sum of the oxidation number of all the atoms in a compound must be zero, and in a polyatomic ion the algebraic sum must equal the charge on the ion, so the sum of three oxygen atoms and one carbon atom in the carbonate ion must equal minus two

What the rules then tell us, and one word about naming

- metallic elements have positive oxidation number and nonmetallic elements have positive or negative oxidation number, while the atoms of transition elements usually display several positive oxidation states
- the highest oxidation number of a representative element is the group number for the first two groups and the group number minus ten, following the long form of periodic table, for the other groups, which implies that the highest value generally increases across the period, changing in the third period from one to seven
- a term often used interchangeably with the oxidation number is the oxidation state, so in carbon dioxide the oxidation state of carbon is plus four, that is also its oxidation number, and similarly for oxygen at minus two, which implies that the oxidation number denotes the oxidation state of an element in a compound

Stock notation, a Roman numeral in brackets

- the oxidation number or state of a metal in a compound is sometimes presented according to the notation given by German chemist Alfred Stock, popularly known as Stock notation, in which the oxidation number is expressed by putting a Roman numeral representing it in parenthesis after the symbol of the metal in the molecular formula
- thus aurous chloride and auric chloride are written as gold one chloride and gold three chloride, and stannous and stannic chloride as tin two chloride and tin four chloride
- this change in oxidation number implies change in oxidation state, which in turn helps to identify whether the species is present in oxidised form or reduced form, so mercury one chloride is the reduced form of mercury two chloride

Problem 7.3, eight compounds in Stock notation

- the question asks us to represent hydrogen tetrachloroaurate, thallium oxide, ferrous oxide, ferric oxide, cuprous iodide, cupric oxide, manganese oxide and manganese dioxide using Stock notation
- by applying various rules the oxidation number of each metallic element is gold three, thallium one, iron two, iron three, copper one, copper two, manganese two and manganese four in turn
- therefore these compounds may be represented with those Roman numerals in parenthesis after the metal symbol in each formula

The five terms restated in oxidation numbers

- oxidation is an increase in the oxidation number of the element in the given substance, and reduction is a decrease in it
- an oxidising agent is a reagent which can increase the oxidation number of an element in a given substance, and these reagents are called oxidants also
- a reducing agent is a reagent which lowers the oxidation number of an element in a given substance, and these reagents are also called reductants
- redox reactions are reactions which involve change in oxidation number of the interacting species

Problem 7.4, cuprous oxide with cuprous sulphide

- the question asks us to justify that two cuprous oxide with cuprous sulphide giving six copper and sulphur dioxide is a redox reaction, and to identify the species oxidised and reduced, which acts as an oxidant and which as a reductant
- assigning oxidation number to each species, we conclude that in this reaction copper is reduced from plus one state to zero oxidation state and sulphur is oxidised from minus two state to plus four state, so the reaction is thus a redox reaction
- the identification of oxidant and reductant is completed overleaf

Group	1	2	13	14	15	16	17
Element	Na	Mg	Al	Si	P	S	Cl
Compound	NaCl	MgSO4	AlF3	SiCl4	P4O10	SF6	HClO
Highest oxidation number state of the group element	+1	+2	+3	+4	+5	+6	+7

Problem 7.4, which is the oxidant

- cuprous oxide helps sulphur in cuprous sulphide to increase its oxidation number, therefore copper one is an oxidant
- sulphur of cuprous sulphide helps copper both in cuprous sulphide itself and in cuprous oxide to decrease its oxidation number, therefore sulphur of cuprous sulphide is reductant

7.3.1 Types of redox reactions

1 Combination reactions

- a combination reaction may be denoted as A plus B giving C, and either A or B or both must be in the elemental form for such a reaction to be a redox reaction
- all combustion reactions which make use of elemental dioxygen, as well as other reactions involving elements other than dioxygen, are redox reactions
- the examples given are carbon with dioxygen giving carbon dioxide, three magnesium with dinitrogen giving magnesium nitride, and methane with two dioxygen giving carbon dioxide and two water

2 Decomposition reactions

- decomposition reactions are the opposite of combination reactions, and precisely a decomposition reaction leads to the breakdown of a compound into two or more components at least one of which must be in the elemental state
- the examples given are two water giving two dihydrogen and dioxygen, two sodium hydride giving two sodium and dihydrogen, and two potassium chlorate giving two potassium chloride and three dioxygen
- it may carefully be noted that there is no change in the oxidation number of hydrogen in methane under combination reactions and that of potassium in potassium chlorate here, and that all decomposition reactions are not redox reactions, the decomposition of calcium carbonate into calcium oxide and carbon dioxide being one that is not

3 Displacement reactions, and metal displacement

- in a displacement reaction an ion or an atom in a compound is replaced by an ion or an atom of another element, denoted as X plus YZ giving XZ plus Y, and displacement reactions fit into two categories, metal displacement and non-metal displacement
- in metal displacement a metal in a compound can be displaced by another metal in the uncombined state, and such reactions find many applications in metallurgical processes in which pure metals are obtained from their compounds in ores
- the examples given are copper sulphate with zinc giving copper and zinc sulphate, vanadium pentoxide with five calcium giving two vanadium and five calcium oxide, titanium tetrachloride with two magnesium giving titanium and two magnesium chloride, and chromium oxide with two aluminium giving aluminium oxide and two chromium
- in each case the reducing metal is a better reducing agent than the one that is being reduced, which evidently shows more capability to lose electrons
- the non-metal displacement redox reactions include hydrogen displacement and a rarely occurring reaction involving oxygen displacement

Metals that displace hydrogen

- all alkali metals and some alkaline earth metals, calcium, strontium and barium, which are very good reductants, will displace hydrogen from cold water, sodium giving sodium hydroxide and dihydrogen and calcium giving calcium hydroxide and dihydrogen
- less active metals such as magnesium and iron react with steam to produce dihydrogen gas, giving magnesium hydroxide and ferric oxide respectively
- many metals, including those which do not react with cold water, are capable of displacing hydrogen from acids, and dihydrogen may even be produced by such metals which do not react with steam, cadmium and tin being examples
- zinc, magnesium and iron with hydrochloric acid give the chloride and dihydrogen, and these reactions are used to prepare dihydrogen gas in the laboratory, the reactivity reflected in the rate of evolution, slowest for the least active metal iron and fastest for the most reactive metal magnesium
- very less active metals which may occur in the native state, such as silver and gold, do not react even with hydrochloric acid

The halogens have an activity series too

- we have already discussed that zinc, copper and silver through tendency to lose electrons show their reducing activity in the order zinc over copper over silver, and like metals an activity series also exists for the halogens
- the power of these elements as oxidising agents decreases as we move down from fluorine to iodine in group 17 of the periodic table, which implies that fluorine is so reactive that it can replace chloride, bromide and iodide ions in solution, and in fact it attacks water and displaces the oxygen of water to give hydrogen fluoride and dioxygen
- it is for this reason that the displacement reactions of chlorine, bromine and iodine using fluorine are not generally carried out in aqueous solution

Chlorine, bromine and the layer test

- chlorine can displace bromide and iodide ions in an aqueous solution, giving potassium chloride with bromine and potassium chloride with iodine, and as bromine and iodine are coloured and dissolve in carbon tetrachloride they can easily be identified from the colour of the solution
- those two reactions can be written in ionic form, and they form the basis of identifying the bromide and iodide ion in the laboratory through the test popularly known as the Layer Test
- bromine likewise can displace iodide ion in solution to give the bromide ion and iodine
- the halogen displacement reactions have a direct industrial application, the recovery of halogens from their halides requiring an oxidation process in which two halide ion gives the halogen plus two electrons, and whereas chemical means are available to oxidise chloride, bromide and iodide, fluorine is the strongest oxidising agent

Why fluorine has to be made electrolytically

- there is no way to convert the fluoride ion to fluorine by chemical means, and the only way to achieve fluorine from the fluoride ion is to oxidise electrolytically, the details of which you will study at a later stage

4 Disproportionation reactions

- disproportionation reactions are a special type of redox reaction in which an element in one oxidation state is simultaneously oxidised and reduced
- one of the reacting substances in a disproportionation reaction always contains an element that can exist in at least three oxidation states, the element in the reacting substance being in the intermediate oxidation state, and both higher and lower oxidation states of that element are formed in the reaction
- the decomposition of hydrogen peroxide is a familiar example, where the oxygen of peroxide, present in the minus one state, is converted to zero oxidation state in dioxygen and decreases to minus two oxidation state in water
- phosphorous, sulphur and chlorine undergo disproportionation in the alkaline medium, white phosphorus with hydroxide giving phosphine and the hypophosphite ion, sulphur with hydroxide giving the sulphide and thiosulphate ions, and chlorine with hydroxide giving the hypochlorite and chloride ions

Bleach, and the one halogen that will not do it

- the chlorine reaction describes the formation of household bleaching agents, the hypochlorite ion formed in it oxidising the colour-bearing stains of the substances to colourless compounds
- whereas bromine and iodine follow the same trend as chlorine, fluorine shows deviation from this behaviour when it reacts with alkali, giving the fluoride ion, oxygen difluoride and water, and it is to be noted with care that fluorine there will undoubtedly attack water to produce some oxygen also
- this departure is not surprising as we know the limitation of fluorine that, being the most electronegative element, it cannot exhibit any positive oxidation state, which means that among halogens fluorine does not show a disproportionation tendency

Problem 7.5, which oxoanion will not disproportionate

- the question lists the hypochlorite, chlorite, chlorate and perchlorate ions and asks which do not show disproportionation and why, and to write the reaction for each that does
- among the oxoanions of chlorine listed, the perchlorate ion does not disproportionate because in it chlorine is present in its highest oxidation state, that is plus seven
- the other three disproportionate, three hypochlorite giving two chloride and one chlorate, six chlorite giving four chlorate and two chloride, and four chlorate giving one chloride and three perchlorate

Problem 7.6, classify four redox reactions

- the question gives four reactions and asks for a scheme of classification, (a) dinitrogen with dioxygen giving two nitric oxide, (b) two lead nitrate giving two lead oxide, four nitrogen dioxide and dioxygen, (c) sodium hydride with water giving sodium hydroxide and dihydrogen, and (d) two nitrogen dioxide with two hydroxide giving the nitrite and nitrate ions and water
- the solution follows on the next page

Problem 7.6, the four classifications

- in reaction (a) the compound nitric oxide is formed by the combination of the elemental substances nitrogen and oxygen, therefore this is an example of combination redox reactions
- reaction (b) involves the breaking down of lead nitrate into three components, therefore this is categorised under decomposition redox reaction
- in reaction (c) hydrogen of water has been displaced by hydride ion into dihydrogen gas, therefore this may be called as displacement redox reaction
- reaction (d) involves disproportionation of nitrogen dioxide in the plus four state into the nitrite ion in the plus three state and the nitrate ion in the plus five state, therefore reaction (d) is an example of disproportionation redox reaction

The paradox of fractional oxidation number

Three compounds where the number comes out a fraction

- sometimes we come across certain compounds in which the oxidation number of a particular element is in fraction, examples being carbon suboxide where the oxidation number of carbon is four thirds, tribromooctaoxide where the oxidation number of bromine is sixteen thirds, and sodium tetrathionate where the oxidation number of sulphur is 2.5
- we know that the idea of fractional oxidation number is unconvincing to us, because electrons are never shared or transferred in fraction, and actually this fractional oxidation state is the average oxidation state of the element under examination
- the structural parameters reveal that the element for whom fractional oxidation state is realised is present in different oxidation states, the structures of carbon suboxide, tribromooctaoxide and the tetrathionate ion each carrying one atom that differs from the rest of the atoms of the same element

What the structures actually show

- in carbon suboxide two carbon atoms are present in the plus two oxidation state each whereas the third is present in zero oxidation state and the average is four thirds, the realistic picture being plus two for two terminal carbons and zero for the middle carbon
- likewise in tribromooctaoxide each of the two terminal bromine atoms is present in plus six oxidation state and the middle bromine in plus four, and once again the average, different from reality, is sixteen thirds
- in the tetrathionate ion each of the two extreme sulphurs exhibits oxidation state of plus five and the two middle sulphurs zero, so the average of the four is 2.5 whereas the reality is plus five, zero, zero and plus five respectively

The conclusion to draw from all this

- the idea of fractional oxidation state should be taken with care and the reality is revealed by the structures only, so whenever we come across a fractional oxidation state of any particular element in any species we must understand that this is the average oxidation number only
- in reality, revealed by structures only, the element in that particular species is present in more than one whole number oxidation states, and ferrous ferric oxide, manganese tetraoxide and red lead are some of the other examples of compounds which are mixed oxides
- however the oxidation states may be in fraction as in the dioxygenyl and superoxide ions, where it is plus one half and minus one half respectively

Problem 7.7, why red lead behaves differently with two acids

- the question asks why red lead with eight hydrochloric acid gives three lead chloride, chlorine and four water, while red lead with four nitric acid gives two lead nitrate, lead dioxide and two water
- red lead is actually a stoichiometric mixture of two mol of lead oxide and one mol of lead dioxide, in which lead is present in plus four oxidation state whereas the stable oxidation state of lead in lead oxide is plus two, so lead dioxide can act as an oxidant and therefore can oxidise the chloride ion of hydrochloric acid into chlorine
- we may also keep in mind that lead oxide is a basic oxide, therefore the first reaction can be splitted into two, an acid-base reaction of lead oxide with hydrochloric acid and a redox reaction of lead dioxide with hydrochloric acid
- since nitric acid itself is an oxidising agent it is unlikely that the reaction may occur between lead dioxide and nitric acid, however the acid-base reaction occurs between lead oxide and nitric acid, and it is the passive nature of lead dioxide against nitric acid that makes the reaction different from the one that follows with hydrochloric acid

7.3.2 Balancing of redox reactions

Two methods, and the first of them

- two methods are used to balance chemical equations for redox processes, one based on the change in the oxidation number of reducing agent and oxidising agent, the other on splitting the redox reaction into two half reactions, one involving oxidation and the other reduction, and the choice of their use rests with the individual using them
- in the oxidation number method, as for other reactions, the compositions and formulas must be known for the substances that react and for the products that are formed

The five steps of the oxidation number method

- step 1, write the correct formula for each reactant and product
- step 2, identify atoms which undergo change in oxidation number by assigning the oxidation number to all elements in the reaction
- step 3, calculate the increase or decrease in the oxidation number per atom and for the entire molecule or ion, and if these are not equal multiply by a suitable number so that they become equal, and if two substances are reduced and nothing is oxidised or the other way about, either the formulas or the oxidation numbers are wrong
- step 4, ascertain the involvement of ions if the reaction is taking place in water and add hydrogen ion or hydroxide ion on the appropriate side so that the total ionic charges of reactants and products are equal, hydrogen ion in acidic solution and hydroxide ion in basic
- step 5, make the numbers of hydrogen atoms on the two sides equal by adding water molecules, then check the number of oxygen atoms, and if these are the same on both sides the equation represents the balanced redox reaction

Problem 7.8, dichromate with sulphite

- the question asks for the net ionic equation for the reaction of potassium dichromate six with sodium sulphite in an acid solution to give the chromium three ion and the sulphate ion, and the solution is worked through overleaf

Problem 7.8, the five steps

- step 1, the skeletal ionic equation is the dichromate ion with the sulphite ion giving the chromium three ion and the sulphate ion
- step 2, assigning oxidation numbers for chromium and sulphur indicates that the dichromate ion is the oxidant and the sulphite ion is the reductant
- step 3, the oxidation state of chromium changes from plus six to plus three, a decrease of three, and that of sulphur from plus four to plus six, an increase of two, so to make the increase and decrease equal place numeral 2 before the chromium ion and numeral 3 before the sulphate ion on the right hand side and balance the chromium and sulphur atoms on both sides
- step 4, as the reaction occurs in the acidic medium and the ionic charges are not equal on both the sides, add eight hydrogen ion on the left to make ionic charges equal
- step 5, finally count the hydrogen atoms and add the appropriate number of water molecules, that is four water, on the right to achieve balanced redox change

Problem 7.9, permanganate with bromide in base

- the question says permanganate ion reacts with bromide ion in basic medium to give manganese dioxide and bromate ion, and asks for the balanced ionic equation
- step 1, the skeletal ionic equation is the permanganate ion with the bromide ion giving manganese dioxide and the bromate ion
- step 2, assigning oxidation numbers for manganese and bromine indicates that permanganate ion is the oxidant and bromide ion is the reductant
- step 3, calculate the increase and decrease of oxidation number and make the increase equal to the decrease, which puts two before the permanganate ion and two before manganese dioxide
- step 4, as the reaction occurs in the basic medium and the ionic charges are not equal on both sides, add two hydroxide ions on the right to make ionic charges equal
- step 5, finally count the hydrogen atoms and add the appropriate number of water molecules, that is one water molecule, on the left side to achieve balanced redox change

The half reaction method, and the first three of its steps

- in this method the two half equations are balanced separately and then added together to give the balanced equation, and it is shown on the oxidation of the ferrous ion to the ferric ion by dichromate ions in acidic medium, wherein dichromate ions are reduced to chromium three ions
- step 1, produce the unbalanced equation for the reaction in ionic form
- step 2, separate the equation into half-reactions, an oxidation half of the ferrous ion giving the ferric ion and a reduction half of the dichromate ion giving the chromium three ion
- step 3, balance the atoms other than oxygen and hydrogen in each half reaction individually, the oxidation half already being balanced with respect to iron atoms while for the reduction half we multiply the chromium three ion by two

Steps 4 to 7 of the half reaction method

- step 4, for reactions occurring in acidic medium add water to balance oxygen atoms and hydrogen ion to balance hydrogen atoms, which puts fourteen hydrogen ion on the left and seven water on the right of the reduction half
- step 5, add electrons to one side of the half reaction to balance the charges, and if need be make the number of electrons equal in the two half reactions by multiplying one or both by an appropriate number, so the oxidation half is rewritten with one electron on the right, and as the reduction half has net twelve positive charges on the left and only six on the right we add six electrons on the left, then multiply the oxidation half by six
- step 6, we add the two half reactions to achieve the overall reaction and cancel the electrons on each side, which gives the net ionic equation
- step 7, verify that the equation contains the same type and number of atoms and the same charges on both sides, and this last check reveals that the equation is fully balanced
- for the reaction in a basic medium first balance the atoms as is done in acidic medium, then for each hydrogen ion add an equal number of hydroxide ions to both sides, and where hydrogen ion and hydroxide ion appear on the same side combine these to give water

Problem 7.10, permanganate with iodide in base

- the question says the permanganate seven ion in basic solution oxidises iodide ion to produce molecular iodine and manganese four oxide, and asks for a balanced ionic equation
- step 1, first we write the skeletal ionic equation, the permanganate ion with the iodide ion giving manganese dioxide and iodine
- step 2, the two half-reactions are an oxidation half of the iodide ion giving iodine and a reduction half of the permanganate ion giving manganese dioxide
- step 3, to balance the iodine atoms in the oxidation half reaction we rewrite it with two iodide ion on the left

Step 4, and why the acid has to be neutralised

- to balance the oxygen atoms in the reduction half reaction we add two water molecules on the right, and to balance the hydrogen atoms we add four hydrogen ion on the left
- as the reaction takes place in a basic solution, therefore for four hydrogen ion we add four hydroxide ions to both sides of the equation
- replacing the hydrogen ion and hydroxide ion with water, the resultant equation is worked out at the top of the next page

Problem 7.10, steps 5 to 7

- the resultant equation is the permanganate ion with two water giving manganese dioxide and four hydroxide ion
- step 5, in this step we balance the charges of the two half-reactions, putting two electrons on the right of the oxidation half and three electrons on the left of the reduction half, then to equalise the number of electrons we multiply the oxidation half-reaction by three and the reduction half-reaction by two
- step 6, add the two half-reactions to obtain the net reaction after cancelling electrons on both sides, giving six iodide with two permanganate and four water to three iodine, two manganese dioxide and eight hydroxide
- step 7, a final verification shows that the equation is balanced in respect of the number of atoms and charges on both sides

7.3.3 Redox reactions as the basis for titrations

(i) and (ii), the reagent as its own indicator

- in acid-base systems we come across a titration method for finding out the strength of one solution against the other using a pH sensitive indicator, and similarly in redox systems the titration method can be adopted to determine the strength of a reductant or oxidant using a redox sensitive indicator
- in one situation the reagent itself is intensely coloured, for example the permanganate ion, which acts as the self indicator, the visible end point achieved after the last of the reductant is oxidised and the first lasting tinge of pink colour appears at a concentration as low as ten to the minus six mol per litre, which ensures a minimal overshoot in colour beyond the equivalence point, the point where the reductant and the oxidant are equal in terms of their mole stoichiometry
- if there is no dramatic auto-colour change there are indicators which are oxidised immediately after the last bit of the reactant is consumed, the best example being the dichromate ion, which is not a self-indicator but oxidises the indicator substance diphenylamine just after the equivalence point to produce an intense blue colour

(iii) The iodine and starch method

- there is yet another method, interesting and quite common, whose use is restricted to those reagents which are able to oxidise iodide ions, for example copper two, where two cupric ion with four iodide give cuprous iodide and iodine
- this method relies on the facts that iodine itself gives an intense blue colour with starch and has a very specific reaction with thiosulphate ions, which too is a redox reaction giving the iodide ion and the tetrathionate ion, and iodine, though insoluble in water, remains in solution containing potassium iodide as the triiodide
- on addition of starch after the liberation of iodine an intense blue colour appears, and this colour disappears as soon as the iodine is consumed by the thiosulphate ions, so the end-point can easily be tracked and the rest is the stoichiometric calculation only

7.3.4 Limitations of the concept, and 7.4 electrode processes

- the concept of redox processes has been evolving with time and this process of evolution is continuing, and in fact in recent past the oxidation process is visualised as a decrease in electron density and reduction as an increase in electron density around the atoms involved in the reaction
- the zinc and copper sulphate experiment can also be observed if a zinc rod is dipped in copper sulphate solution, where the redox reaction takes place and during the reaction the account continues on the next page

Two beakers, zinc in zinc sulphate and copper in copper sulphate

A salt bridge joining the solutions, with a wire, ammeter and switch above

Electrons leaving the zinc anode, crossing the wire, reducing copper ions at the cathode

Figure 7.C The set-up for Daniell cell. Electrons produced at the anode due to oxidation of Zn travel through the external circuit to the cathode where these reduce the copper ions. The circuit is completed inside the cell by the migration of ions through the salt bridge. It may be noted that the direction of current is opposite to the direction of electron flow.

Separating the two halves, and the redox couple

- zinc is oxidised to zinc ions and copper ions are reduced to metallic copper due to direct transfer of electrons from zinc to copper ion, and during this reaction heat is also evolved
- now we modify the experiment so that for the same redox reaction transfer of electrons takes place indirectly, which necessitates the separation of zinc metal from copper sulphate solution, taking copper sulphate solution in a beaker with a copper strip in it and zinc sulphate solution in another beaker with a zinc rod in it
- at the interface of the metal and its salt solution in each beaker both the reduced and oxidized forms of the same species are present, and a redox couple is defined as having together the oxidised and reduced forms of a substance taking part in an oxidation or reduction half reaction
- this is represented by separating the oxidised form from the reduced form by a vertical line or a slash representing an interface, so the two couples here are written with the oxidised form put before the reduced form in both cases

The salt bridge, and what happens at the switch

- we connect solutions in the two beakers by a salt bridge, a U-tube containing a solution of potassium chloride or ammonium nitrate usually solidified by boiling with agar agar and later cooling to a jelly like substance, which provides an electric contact between the two solutions without allowing them to mix
- the zinc and copper rods are connected by a metallic wire with a provision for an ammeter and a switch, and the set-up is known as Daniell cell
- when the switch is in the off position no reaction takes place in either beaker and no current flows, and as soon as it is on, first, the transfer of electrons now does not take place directly from zinc to the copper ion but through the metallic wire connecting the two rods, and second, the electricity from solution in one beaker to the other flows by the migration of ions through the salt bridge

Electrode potential, and the standard state

- the flow of current is possible only if there is a potential difference between the copper and zinc rods, known as electrodes here, and the potential associated with each electrode is known as electrode potential
- if the concentration of each species taking part in the electrode reaction is unity, any gas appearing being confined to 1 atmospheric pressure, and further the reaction is carried out at 298K, then the potential of each electrode is said to be the Standard Electrode Potential, and by convention that of the hydrogen electrode is 0.00 volts
- the electrode potential value for each electrode process is a measure of the relative tendency of the active species in the process to remain in the oxidised or reduced form, a negative value meaning that the redox couple is a stronger reducing agent than the hydrogen ion over dihydrogen couple

What the sign of the value tells you

- a positive standard electrode potential means that the redox couple is a weaker reducing agent than the hydrogen ion over dihydrogen couple, and a negative one that it is a stronger reducing agent than that couple
- the standard electrode potentials are very important and we can get a lot of other useful information from them, and the values for some selected electrode processes, which are reduction reactions, are given in the table below
- the table runs from fluorine at 2.87 volts, the strongest oxidising agent in the list, down to lithium at minus 3.05 volts, the strongest reducing agent, so oxidising strength increases up the table and reducing strength down it, and you will learn more about electrode reactions and cells in Class XII

Table 7.A The Standard Electrode Potentials at 298 K

Oxidised form + ne–	Reduced form	E / V
F2(g) + 2e–	2F–	2.87
Co3+ + e–	Co2+	1.81
H2O2 + 2H+ + 2e–	2H2O	1.78
MnO4– + 8H+ + 5e–	Mn2+ + 4H2O	1.51
Au3+ + 3e–	Au(s)	1.40
Cl2(g) + 2e–	2Cl–	1.36
Cr2O72– + 14H+ + 6e–	2Cr3+ + 7H2O	1.33
O2(g) + 4H+ + 4e–	2H2O	1.23
MnO2(s) + 4H+ + 2e–	Mn2+ + 2H2O	1.23
Br2 + 2e–	2Br–	1.09
NO3– + 4H+ + 3e–	NO(g) + 2H2O	0.97
2Hg2+ + 2e–	Hg22+	0.92
Ag+ + e–	Ag(s)	0.80
Fe3+ + e–	Fe2+	0.77
O2(g) + 2H+ + 2e–	H2O2	0.68
I2(s) + 2e–	2I–	0.54
Cu+ + e–	Cu(s)	0.52
Cu2+ + 2e–	Cu(s)	0.34
AgCl(s) + e–	Ag(s) + Cl–	0.22
AgBr(s) + e–	Ag(s) + Br–	0.10
2H+ + 2e–	H2(g)	0.00
Pb2+ + 2e–	Pb(s)	–0.13
Sn2+ + 2e–	Sn(s)	–0.14
Ni2+ + 2e–	Ni(s)	–0.25
Fe2+ + 2e–	Fe(s)	–0.44
Cr3+ + 3e–	Cr(s)	–0.74
Zn2+ + 2e–	Zn(s)	–0.76
2H2O + 2e–	H2(g) + 2OH–	–0.83
Al3+ + 3e–	Al(s)	–1.66
Mg2+ + 2e–	Mg(s)	–2.36
Na+ + e–	Na(s)	–2.71
Ca2+ + 2e–	Ca(s)	–2.87
K+ + e–	K(s)	–2.93
Li+ + e–	Li(s)	–3.05

Ions are present as aqueous species and H2O as liquid; gases and solids are shown by g and s respectively.

Summary

What this unit covered

- redox reactions form an important class of reactions in which oxidation and reduction occur simultaneously, and three tier conceptualisation, classical, electronic and oxidation number, which is usually available in the texts, has been presented in detail
- oxidation, reduction, oxidising agent and reducing agent have been viewed according to each conceptualisation, and oxidation numbers are assigned in accordance with a consistent set of rules
- oxidation number and ion-electron method both are useful means in writing equations for the redox reactions, and redox reactions are classified into four categories, combination, decomposition displacement and disproportionation reactions
- the concept of redox couple and electrode processes is introduced here, and the redox reactions find wide applications in the study of electrode processes and cells

Exercises 7.1 to 7.9

Exercises 7.1 to 7.5

- 7.1 Assign oxidation number to the underlined elements in each of the following species: (a) NaH2PO4 (b) NaHSO4 (c) H4P2O7 (d) K2MnO4 (e) CaO2 (f) NaBH4 (g) H2S2O7 (h) KAl(SO4)2.12 H2O
- 7.2 What are the oxidation number of the underlined elements in each of the following and how do you rationalise your results? (a) KI3 (b) H2S4O6 (c) Fe3O4 (d) CH3CH2OH (e) CH3COOH
- 7.3 Justify that the following reactions are redox reactions: (a) CuO(s) + H2(g) giving Cu(s) + H2O(g) (b) Fe2O3(s) + 3CO(g) giving 2Fe(s) + 3CO2(g) (c) 4BCl3(g) + 3LiAlH4(s) giving 2B2H6(g) + 3LiCl(s) + 3 AlCl3 (s) (d) 2K(s) + F2(g) giving 2K+F- (s) (e) 4 NH3(g) + 5 O2(g) giving 4NO(g) + 6H2O(g)
- 7.4 Fluorine reacts with ice and results in the change H2O(s) + F2(g) giving HF(g) + HOF(g). Justify that this reaction is a redox reaction.
- 7.5 Calculate the oxidation number of sulphur, chromium and nitrogen in H2SO5, Cr2O72- and NO3-. Suggest structure of these compounds. Count for the fallacy.

Exercises 7.6 to 7.9

- 7.6 Write formulas for the following compounds: (a) Mercury(II) chloride (b) Nickel(II) sulphate (c) Tin(IV) oxide (d) Thallium(I) sulphate (e) Iron(III) sulphate (f) Chromium(III) oxide
- 7.7 Suggest a list of the substances where carbon can exhibit oxidation states from -4 to +4 and nitrogen from -3 to +5.
- 7.8 While sulphur dioxide and hydrogen peroxide can act as oxidising as well as reducing agents in their reactions, ozone and nitric acid act only as oxidants. Why?
- 7.9 Consider the reactions: (a) 6 CO2(g) + 6H2O(l) giving C6 H12 O6(aq) + 6O2(g) (b) O3(g) + H2O2(l) giving H2O(l) + 2O2(g). Why it is more appropriate to write these reactions as (a) 6CO2(g) + 12H2O(l) giving C6 H12 O6(aq) + 6H2O(l) + 6O2(g) (b) O3(g) + H2O2 (l) giving H2O(l) + O2(g) + O2(g). Also suggest a technique to investigate the path of the above (a) and (b) redox reactions.

Exercises, continued

Exercises 7.10 to 7.13

- 7.10 The compound AgF2 is unstable compound. However, if formed, the compound acts as a very strong oxidising agent. Why?
- 7.11 Whenever a reaction between an oxidising agent and a reducing agent is carried out, a compound of lower oxidation state is formed if the reducing agent is in excess and a compound of higher oxidation state is formed if the oxidising agent is in excess. Justify this statement giving three illustrations.
- 7.12 How do you count for the following observations? (a) Though alkaline potassium permanganate and acidic potassium permanganate both are used as oxidants, yet in the manufacture of benzoic acid from toluene we use alcoholic potassium permanganate as an oxidant. Why? Write a balanced redox equation for the reaction. (b) When concentrated sulphuric acid is added to an inorganic mixture containing chloride, we get colourless pungent smelling gas HCl, but if the mixture contains bromide then we get red vapour of bromine. Why?
- 7.13 Identify the substance oxidised reduced, oxidising agent and reducing agent for each of the following reactions: (a) 2AgBr (s) + C6H6O2(aq) giving 2Ag(s) + 2HBr (aq) + C6H4O2(aq) (b) HCHO(l) + 2[Ag (NH3)2]+(aq) + 3OH-(aq) giving 2Ag(s) + HCOO-(aq) + 4NH3(aq) + 2H2O(l) (c) HCHO (l) + 2 Cu2+ (aq) + 5 OH- (aq) giving Cu2O(s) + HCOO- (aq) + 3H2O(l) (d) N2H4(l) + 2H2O2(l) giving N2(g) + 4H2O(l) (e) Pb(s) + PbO2(s) + 2H2SO4(aq) giving 2PbSO4(s) + 2H2O(l)

Exercises 7.14 to 7.17

- 7.14 Consider the reactions 2 S2O32- (aq) + I2(s) giving S4 O62-(aq) + 2I-(aq), and S2O32-(aq) + 2Br2(l) + 5 H2O(l) giving 2SO42-(aq) + 4Br-(aq) + 10H+(aq). Why does the same reductant, thiosulphate react differently with iodine and bromine?
- 7.15 Justify giving reactions that among halogens, fluorine is the best oxidant and among hydrohalic compounds, hydroiodic acid is the best reductant.
- 7.16 Why does the following reaction occur? XeO64- (aq) + 2F- (aq) + 6H+(aq) giving XeO3(g) + F2(g) + 3H2O(l). What conclusion about the compound Na4XeO6, of which XeO64- is a part, can be drawn from the reaction.
- 7.17 Consider the reactions: (a) H3PO2(aq) + 4 AgNO3(aq) + 2 H2O(l) giving H3PO4(aq) + 4Ag(s) + 4HNO3(aq) (b) H3PO2(aq) + 2CuSO4(aq) + 2 H2O(l) giving H3PO4(aq) + 2Cu(s) + H2SO4(aq) (c) C6H5CHO(l) + 2[Ag (NH3)2]+(aq) + 3OH-(aq) giving C6H5COO-(aq) + 2Ag(s) + 4NH3 (aq) + 2 H2O(l) (d) C6H5CHO(l) + 2Cu2+ (aq) + 5OH- (aq) giving No change observed.

Exercises, continued

Exercises 7.18 to 7.21

- the page opens by finishing 7.17, asking what inference you draw about the behaviour of Ag+ and Cu2+ from those four reactions
- 7.18 Balance the following redox reactions by ion - electron method: (a) MnO4- (aq) + I- (aq) giving MnO2 (s) + I2(s) (in basic medium) (b) MnO4- (aq) + SO2 (g) giving Mn2+ (aq) + HSO4- (aq) (in acidic solution) (c) H2O2 (aq) + Fe2+ (aq) giving Fe3+ (aq) + H2O (l) (in acidic solution) (d) Cr2O72- + SO2(g) giving Cr3+ (aq) + SO42- (aq) (in acidic solution)
- 7.19 Balance the following equations in basic medium by ion-electron method and oxidation number methods and identify the oxidising agent and the reducing agent: (a) P4(s) + OH-(aq) giving PH3(g) + HPO2- (aq) (b) N2H4(l) + ClO3-(aq) giving NO(g) + Cl-(g) (c) Cl2O7 (g) + H2O2(aq) giving ClO2-(aq) + O2(g) + H+
- 7.20 What sorts of informations can you draw from the following reaction? (CN)2(g) + 2OH-(aq) giving CN-(aq) + CNO-(aq) + H2O(l)
- 7.21 The Mn3+ ion is unstable in solution and undergoes disproportionation to give Mn2+, MnO2, and H+ ion. Write a balanced ionic equation for the reaction.

Exercises 7.22 to 7.26

- 7.22 Consider the elements Cs, Ne, I and F. (a) Identify the element that exhibits only negative oxidation state. (b) Identify the element that exhibits only postive oxidation state. (c) Identify the element that exhibits both positive and negative oxidation states. (d) Identify the element which exhibits neither the negative nor does the positive oxidation state.
- 7.23 Chlorine is used to purify drinking water. Excess of chlorine is harmful. The excess of chlorine is removed by treating with sulphur dioxide. Present a balanced equation for this redox change taking place in water.
- 7.24 Refer to the periodic table given in your book and now answer the following questions: (a) Select the possible non metals that can show disproportionation reaction. (b) Select three metals that can show disproportionation reaction.
- 7.25 In Ostwald's process for the manufacture of nitric acid, the first step involves the oxidation of ammonia gas by oxygen gas to give nitric oxide gas and steam. What is the maximum weight of nitric oxide that can be obtained starting only with 10.00 g. of ammonia and 20.00 g of oxygen?
- 7.26 Using the standard electrode potentials given in the Table 8.1, predict if the reaction between the following is feasible: (a) Fe3+(aq) and I-(aq) (b) Ag+(aq) and Cu(s) (c) Fe3+ (aq) and Cu(s) (d) Ag(s) and Fe3+(aq) (e) Br2(aq) and Fe2+(aq).

Two things printed here as they stand

- 7.22 (b) prints postive for positive, and it is carried here as printed
- 7.26 sends the reader to Table 8.1 for the standard electrode potentials, but the table of standard electrode potentials in this unit is Table 7.1, printed eight pages earlier

Exercises, continued

Exercises 7.27 and 7.28

- 7.27 Predict the products of electrolysis in each of the following: (i) An aqueous solution of AgNO3 with silver electrodes (ii) An aqueous solution AgNO3 with platinum electrodes (iii) A dilute solution of H2SO4 with platinum electrodes (iv) An aqueous solution of CuCl2 with platinum electrodes.
- 7.28 Arrange the following metals in the order in which they displace each other from the solution of their salts: Al, Cu, Fe, Mg and Zn.

Exercises 7.29 and 7.30

- 7.29 Given the standard electrode potentials K+/K = -2.93V, Ag+/Ag = 0.80V, Hg2+/Hg = 0.79V, Mg2+/Mg = -2.37V, Cr3+/Cr = -0.74V, arrange these metals in their increasing order of reducing power.
- 7.30 Depict the galvanic cell in which the reaction Zn(s) + 2Ag+(aq) giving Zn2+(aq) +2Ag(s) takes place. Further show (i) which of the electrode is negatively charged, (ii) the carriers of the current in the cell, and (iii) individual reaction at each electrode.